

“Math is Cool” Championships -- 2025-26

4th Grade

Mental Math Solutions

	Answer	Solution
1	8 [dragons]	Slade can slay two dragons in one minute. How many dragons can he slay in four minutes? $2 \times 4 = 8$
2	12 [quarters]	How many quarters does it take to make three dollars? 1 dollar = 4 quarters 3 dollars = 12 quarters
3	64 [sq. feet]	What is the area in square feet of the largest square that can fit into a rectangle of dimensions eight feet by twelve feet? $8 \times 8 = 64$
4	32	A sequence starts as follows: eight, fourteen, twenty, twenty-six, and so on. What is the next number in the sequence? Add 6 each time: 8, 14, 20, 26, 32
5	10 [times]	A broken clock chimes once at the top of every hour. How many times does it chime between nine forty-three PM and seven twelve AM the next morning? 10, 11, 12, 1, 2, 3, 4, 5, 6, 7
6	60 [= smallest integer]	What is the smallest integer greater than two point nine nine times three point nine nine times four point nine nine? Estimate: $3 \times 4 \times 5 = 60$
7	25 [%]	Gracie has a bag with three red marbles, five blue marbles, and twelve green marbles. She picks one marble from the bag without looking. What is the probability in percent that she picks a blue marble? 5 blue marbles out of 20 total marbles = 25%

8

14 [%]

What percent of the whole numbers from one to one hundred are prime numbers that contain the digit three or the digit seven but not both?

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

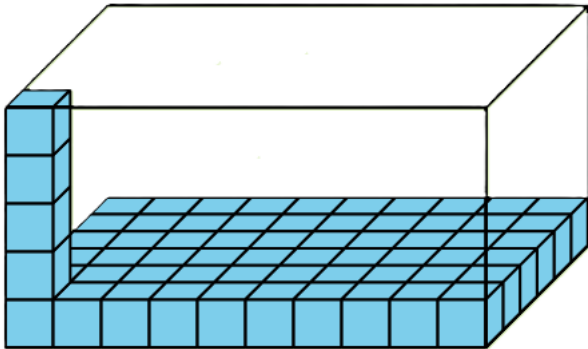
“Math is Cool” Championships -- 2025-26

4th Grade

Individual Test Solutions

	Answer	Solution											
1	35 [pages]	Gabbie reads 5 pages every day. How many pages will she read in 7 days? $7 \times 5 = 35$											
2	24 [cookies]	How many cookies are in 2 dozen cookies? $2 \times 12 = 24$											
3	3 [socks]	Thirteen red socks and 13 blue socks are in a drawer. What is the smallest number of socks Owynn needs to take out of the drawer without looking to ensure there are at least two socks of the same color? She can pull out one blue and then one red, and then the third sock must be the same as one of the previous.											
4	1050	What is 4,500 minus 3,450? $4500 - 3,450 = 1050$											
5	24	A whole number is greater than 20 and less than 30. It is divisible by both 2 and 3. What number is it? 24 is the only number between 20 and 30 divisible by 6, or 3×2 .											
6	10 [\$]	There are 5 toy cars on a shelf. Their prices (in dollars) are: \$10, \$12, \$8, \$10, \$10 What is the average price of the toy cars in dollars? The sum is 50. $50 / 5 = 10$											
7	55 [apples]	The number of apples picked by students is shown in the table. How many apples were picked in total? Total = $10 + 16 + 21 + 8 = 55$	<table><tr><th>Student</th><th>Number of apples</th></tr><tr><td>Aadi</td><td>10</td></tr><tr><td>Kepler</td><td>16</td></tr><tr><td>Yuvaan</td><td>21</td></tr><tr><td>Diane</td><td>8</td></tr></table>	Student	Number of apples	Aadi	10	Kepler	16	Yuvaan	21	Diane	8
Student	Number of apples												
Aadi	10												
Kepler	16												
Yuvaan	21												
Diane	8												
8	2 [cupcakes]	Athena has 23 cupcakes. Each plate can hold 3 cupcakes. If she fills up as many plates with cupcakes as she can, how many cupcakes are left over? 21 is divisible by 3 $23 - 21 = 2$ cupcakes											
9	29 [= x]	Solve for x: $x - 6 = 23$ $23 + 6 = 29$											
10	22 [units]	A rectangle has a length of 8 units and a width of 3 units. What is the perimeter of the rectangle in units? $8 + 8 + 3 + 3 = 22$											

11	90 [= median]	Here are the scores from a spelling test: 88, 92, 85, 90, 95 What is the median score? 90 is the middle number, or median.
12	20 [more consonants]	If Triston writes "MATHISCOOL" 10 times, how many more total consonants than total vowels does he write? 6 consonants - 4 vowels per write-out = 2 more consonants 2 consonants x 10 times = 20 more consonants
13	20 [years]	Yanek is 7 years old and his mother is 34. After how many years will Yanek be half his mother's age? Yanek's mother was 27 when he was born. When Yanek is 27, he will be half her age. He is already 7, so he will be 27 in 20 years.
14	70 [%]	There are 14 apples and 6 oranges in a bag. If one fruit is randomly taken out of the bag, what is the probability in percent that it is an apple? $14 \times (100/20) = 70$
15	25 [prisms]	A pile of clay can be molded into a cube with side lengths of 10 inches. Using all the clay, how many rectangular prisms with dimensions 2 x 4 x 5 inches can be made? 1000 cubic inches of clay 40 cubic inches per prism $1000 / 40 = 25$ prisms
16	16 [whole numbers]	How many whole numbers are there between 7 and 24, not counting 7 and 24? $24 - 7 - 1 = 16$
17	6 [2s]	In this image of 10-sided dice, how many sides numbered with a 2 are visible? 
18	31	What is the next number in this sequence? 6, 11, 16, 21, 26, ... The pattern is +5 every time.

19	275 [cents]	Two dollars, 1 quarter, 3 dimes and 4 nickels equals how many total cents? $200 + 25 + 30 + 20 = 275$ cents
20	105 [miles]	A car travels 63 miles in 3 hours. At the same rate, how many miles does the car travel in 5 hours? $63 / 3 = 21$ miles in 1 hour $21 * 5 = 105$
21	19	Evaluate, and answer as a whole number: $9.2 + 11.7 - 1.9$ $9.2 + 11.7 - 1.9 = 19$
22	90	A number pattern starts like this: 2, 6, 12, 20, 30 . . . What is the 9 th number in this sequence? Pattern is +4, +6, +8, +10, etc. 2, 6, 12, 20, 30, 42, 56, 72, 90
23	8 [8 th]	There are 29 students competing in a race. The number of students that finished behind Weibo is 3 times as large as the number that finished ahead of Weibo. In which place did Weibo finish? 28 other contestants. 21 behind Grant equals three times the 7 ahead of him.
24	236 [unit cubes]	This rectangular prism is being filled with unit cubes. How many more unit cubes are needed to fill the prism?  The dimensions are $5 \times 6 \times 10 = 300$ total unit cubes needed. There are currently $10 \times 6 + 4 = 64$ cubes. $300 - 64 = 236$ more are needed.
25	18 [days]	Every day, a cafeteria offers the choice of a corn dog, nachos, or pizza as lunch entrees. This comes with the choices of water, milk, or soda to drink and a cookie or pie for dessert. If a different combination of one entree, one drink, and one dessert is tried every day, how many days does it take to try every combination? $3 \text{ entree choices} \times 3 \text{ drink choices} \times 2 \text{ dessert choices} = 18$ possible combinations
26	95 [\$]	Harini buys 3 markers that all cost the same and 2 erasers that cost \$2 each. He spends \$19 in total. How many dollars would 19 markers cost? $19 - (2 \times 2) = 15$ $15 / 3 = 5$ $5 * 19 = 95$

27	13 [squares]	A rectangle has side lengths of 19 units and 47 units. Eloise cuts it to obtain the largest possible square and another rectangle. She then continues to cut the remaining rectangle in the same way until there is no rectangle left. How many squares does she have at the end? 2 squares of 19 x 19 2 squares of 9 x 9 9 squares of 1 x 1																																																																																																																																		
28	16 [balls]	Katarina has a bag with 13 red balls and 7 blue balls. Without looking, how many balls does Katarina have to pick before she can be certain to pick at least 9 red balls? She could pick 7 blue balls first and then 9 red balls, equaling 16 balls.																																																																																																																																		
29	154	A shift cipher is a way to write a secret code by shifting each letter in the alphabet by the same amount. If the end of the alphabet is reached, then go back to the start at letter A. Using a shift cipher, the word MATH is encoded as SGZN. As a whole number, what does the following code represent? UTK NATJXKJ LOLZE-LUAX <table><tr><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td><td>H</td><td>I</td><td>J</td><td>K</td><td>L</td><td>M</td><td>N</td><td>O</td><td>P</td><td>Q</td><td>R</td><td>S</td><td>T</td><td>U</td><td>V</td><td>W</td><td>X</td><td>Y</td><td>Z</td></tr><tr><td>G</td><td>H</td><td>I</td><td>J</td><td>K</td><td>L</td><td>M</td><td>N</td><td>O</td><td>P</td><td>Q</td><td>R</td><td>S</td><td>T</td><td>U</td><td>V</td><td>W</td><td>X</td><td>Y</td><td>Z</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>O</td><td>N</td><td>E</td><td>H</td><td>U</td><td>N</td><td>D</td><td>R</td><td>E</td><td>D</td><td>F</td><td>I</td><td>F</td><td>T</td><td>Y</td><td>-</td><td>F</td><td>O</td><td>U</td><td>R</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>U</td><td>T</td><td>K</td><td>N</td><td>A</td><td>T</td><td>J</td><td>X</td><td>K</td><td>J</td><td>L</td><td>O</td><td>L</td><td>Z</td><td>E</td><td>-</td><td>L</td><td>U</td><td>A</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F																											O	N	E	H	U	N	D	R	E	D	F	I	F	T	Y	-	F	O	U	R							U	T	K	N	A	T	J	X	K	J	L	O	L	Z	E	-	L	U	A	X						
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30	75 [%]	A spinner is divided into 8 equal parts numbered 1 to 8. On a single spin, what is the probability in percent of landing on an even number, or a number greater than 4? 2, 4, 5, 6, 7, 8 is 6 numbers out of 8, $6/8 = \frac{3}{4} = 75\%$																																																																																																																																		
31	8 [cards]	Kingston has 16 cards. Each card is labeled with a different number from 1 to 16. What is the largest number of cards Kingston can discard if he wants to keep a set of cards that add up to at least 100? $16+15+14+13+12+11+10+9= 100$																																																																																																																																		
32	[A =] 27	A cylinder has a radius of 15 millimeters and a height of 0.12 meters. In terms of π and in cubic centimeters, the volume of the prism can be expressed as $A\pi$ cm^3 . What is the value of A? $15 \text{ mm} = 1.5 \text{ cm}$ and $0.12 \text{ m} = 12 \text{ cm}$, so volume = $(1.5)^2\pi \cdot 12 = 2.25\pi \cdot 12 = 27\pi$ cm^3 and $A = 27$.																																																																																																																																		
33	8 [or -8, or both]	Taylee has a number. She multiplies it by itself and then adds 8 before dividing by 12. If she ends up with 6, what number did she start with? $6 * 12 = 72$ $72 - 8 = 64$ Square root of 64 is 8 Correction: question should have said a POSITIVE number. Both answers accepted.																																																																																																																																		

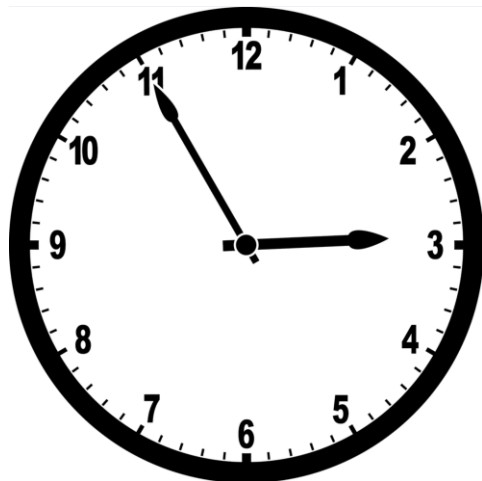
34	11 [= the other factor]	A number is between 40 and 60. It has exactly four factors: 1, itself, and two other numbers. One of those factors is 5. What is the other factor? The only numbers between 40 and 60 with a factor of 5 are 45, 50, and 55. 45 and 50 have more than four factors, so it must be 55. $55 / 5 = 11$
35	15 [= sum]	The pyramid is an example of a "self-descriptive" sequence, where each row describes the digits that are in the previous row. For example, the 2nd row tells us that there was one '2' in the first row. The 3rd row tells us that there was one '1' and one '2' in the second row. A total of 5 rows of the pyramid are included. What is the sum of the digits in the 6th row? 2 1 2 1 1 1 2 3 1 1 2 1 3 2 1 1 2 $1 + 1 + 1 + 3 + 1 + 2 + 2 + 1 + 1 + 2 = 15$ 2 1 2 1 1 1 2 3 1 1 2 1 3 2 1 1 2 1 1 1 3 1 2 2 1 1 2
36	[R =] 22	A set of five distinct positive integers has a mean of 9, which is also the median. What is the largest possible range, R, where R = the largest integer minus the smallest integer in the set. The five integers must add up to $5 \cdot 9 = 45$, and the middle number is 9. The set with the maximum range is 1, 2, 9, 10, 23, so $R = 23 - 1 = 22$.
37	252 [3-digit integers]	How many positive three-digit integers contain the digit 8 at least once? Case study: Three 8's: 1 way (888) Two 8's: 26 ways (9 ways for 88_, 9 ways for 8_8, 8 ways for _88) One 8: 225 ways (81 ways for 8_-, 72 ways for _8-, 72 ways for __8). $1 + 26 + 225 = 252$

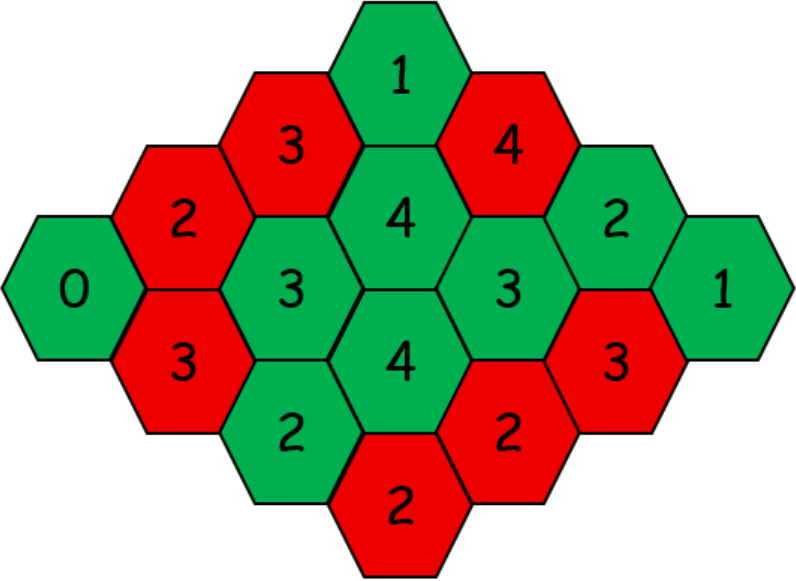
38	175 [nickels]	There are 1000 coins in a jar: pennies, nickels, dimes and quarters. The probability of reaching in and randomly selecting a quarter is 3/5. The probability of reaching in and randomly selecting a dime is 1/8. Ten percent of the coins are pennies. How many nickels are in the jar? 3/5(1000) = 600 quarters 1/8(1000) = 125 dimes 10% of 1000 = 100 1000 - 600 - 125 - 100 = 175 nickels																																																	
39	11 [= sum]	In the following puzzle, place the digits 1 through 9 in the heavily outlined squares, using each digit exactly once, in order to make the three horizontal and three vertical equations true. Calculations are done from left to right and top to bottom, NOT using the usual "order of operations". For example: 3 + 4 x 2 = 7 x 2 = 14. What is the sum of the two numbers indicated by question marks? <table><tr><td>7</td><td>-</td><td>6</td><td>+</td><td>8</td><td>=</td><td>9</td></tr><tr><td>+</td><td></td><td>+</td><td></td><td>-</td><td></td><td></td></tr><tr><td>1</td><td>+</td><td>5</td><td>x</td><td>3</td><td>=</td><td>18</td></tr><tr><td>+</td><td></td><td>+</td><td></td><td>-</td><td></td><td></td></tr><tr><td>9</td><td>+</td><td>4</td><td>x</td><td>2</td><td>=</td><td>26</td></tr><tr><td>=</td><td></td><td>=</td><td></td><td>=</td><td></td><td></td></tr><tr><td>17</td><td></td><td>15</td><td></td><td>3</td><td></td><td></td></tr></table>	7	-	6	+	8	=	9	+		+		-			1	+	5	x	3	=	18	+		+		-			9	+	4	x	2	=	26	=		=		=			17		15		3		
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40	11 [cups]	Xander builds a pyramid of cups with 13 in the bottom row, 12 in the second row, 11 in the third row, and so on, until a single cup stands on top. A gust of wind blows down the top 5 rows of cups and half of the sixth row from the top. With the remaining number of standing cups, Xander builds the largest pyramid possible. How many cups are in the bottom row of the new pyramid? 13 + 12 . . . + 2 + 1 = 91 cups 1 + 2 + 3 + 4 + 5 + 3 = 18 cups lost. This means the row of 13 and row of 12 are both incomplete, so the bottom row must be 11 cups.																																																	

"Math is Cool" Championships -- 2025-26

4th Grade

Multiple Choice Solutions

	Answer	Solution
USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #1 THROUGH #3. The clock is currently showing the time in the afternoon of Friday, February 13, 2026. 		
1	B	What time will the clock show in 135 minutes? A) 4:40 pm B) 5:10 pm C) 5:40 pm D) 6:10 pm E) Answer not given. 135 minutes is 2 hours 15 minutes. 2 hours is 4:55 and 15 minutes is 5:10
2	D	Including the number it starts on (11), how many total numbers on the clock face will the minute hand pass to reach the time of 9:42 pm on February 13, 2026? A) 79 B) 80 C) 81 D) 82 E) Answer not given. Starting at 2:55, on the 11, it passes 12 numbers to get to 3:54. And so on, it was pass 12 numbers each hour, so from 2:55 to 8:54 is 6 hours, $6 \times 12 = 72$. From 8:55 to 9:42 it passes 10 numbers (11, 12, 1, 2, 3, 4, 5, 6, 7, 8). Total = 82.

6	D	How many of the hexagons contain marbles? A) 6 B) 7 C) 8 D) 9 E) None of the above. Work from left to right, knowing that the left-most hexagon does not have any touching hexagons that contain marbles. Green = has marbles, red = no marbles. 
7	C	Out of the hexagons that contain marbles, at least one contains only one marble, at least two contain an even number of marbles, and the maximum number of marbles in any one hexagon is five. The sum of all of the marbles is divisible by 4. What is the maximum possible number of hexagons that contain five marbles? A) 3 B) 4 C) 5 D) 6 E) None of the above. A possible starting point with one 1, two evens, and the rest fives is: 1 4 4 5 5 5 5 5 5. Sum = 39, so we need to get down to 36 to be divisible by 4. It is impossible to do this by keeping 6 fives, so take away one of the fives: 1 2 4 4 5 5 5 5 5. Sum = 36. Therefore the maximum is 5 fives.

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #8 THROUGH #10.

A number of people were surveyed about their favorite pizza toppings, with the choices being: cheese, pepperoni, sausage, olives, and mushrooms. Each person picked one favorite topping. The results of the survey are as follows:

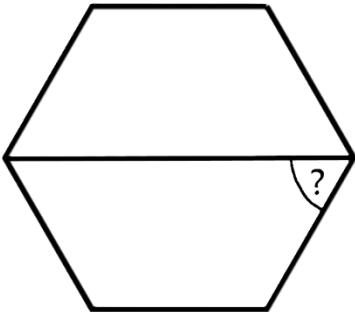
- Thirty-six people said that olives were their favorite topping.
- The number of people who picked olives is 12 more than the number of people who said that mushrooms are their favorite topping.
- The ratio of people who picked mushrooms to those who picked pepperoni is 1:3.
- The number of people who picked sausage is 12 more than the number who picked olives as their favorite.
- 25% of the people picked cheese as their favorite topping

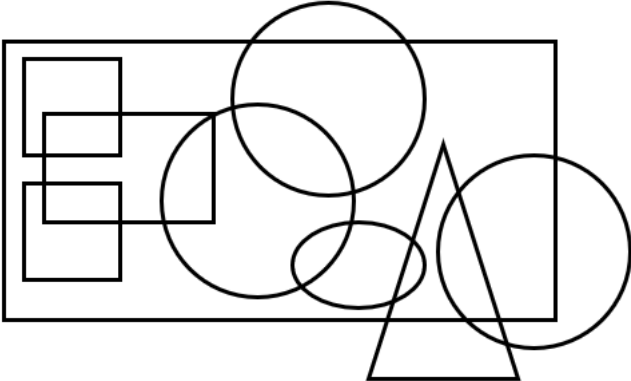
8	A	How many people said that mushrooms are their favorite topping? A) 24 B) 36 C) 48 D) 52 E) None of the above. Olives = 36 = 12 more than mushrooms. Mushrooms = 36 - 12 = 24															
9	D	How many total people were surveyed? A) 120 B) 180 C) 220 D) 240 E) None of the above. <table><tr><th>Olives</th><th>Mushrooms</th><th>Pepperoni</th><th>Sausage</th><th>Cheese</th></tr><tr><td>Given</td><td>$M = O - 12 = 24$</td><td>M: P is 1 : 3 $P = 3M = 3(24) = 72$</td><td>$S = O + 12 = 36 + 12 = 48$</td><td>Cheese = 25% of total Rest = 75% of total $36 + 24 + 72 + 48 = 180 = 3/4T$ $T = 180(4/3) = 240$ Cheese = $240 - 180 = 60$</td></tr><tr><td>36</td><td>24</td><td>72</td><td>48</td><td>60</td></tr></table>	Olives	Mushrooms	Pepperoni	Sausage	Cheese	Given	$M = O - 12 = 24$	M: P is 1 : 3 $P = 3M = 3(24) = 72$	$S = O + 12 = 36 + 12 = 48$	Cheese = 25% of total Rest = 75% of total $36 + 24 + 72 + 48 = 180 = 3/4T$ $T = 180(4/3) = 240$ Cheese = $240 - 180 = 60$	36	24	72	48	60
Olives	Mushrooms	Pepperoni	Sausage	Cheese													
Given	$M = O - 12 = 24$	M: P is 1 : 3 $P = 3M = 3(24) = 72$	$S = O + 12 = 36 + 12 = 48$	Cheese = 25% of total Rest = 75% of total $36 + 24 + 72 + 48 = 180 = 3/4T$ $T = 180(4/3) = 240$ Cheese = $240 - 180 = 60$													
36	24	72	48	60													
10	B	A Medium Cheese pizza costs \$10.00, and additional toppings cost \$1.20 each. Biff and Eho order a Medium Cheese pizza with pepperoni, sausage and mushrooms. If they divide the cost evenly, how much does each of them pay? A) \$3.40 B) \$6.80 C) \$7.20 D) \$13.60 E) None of the above. $10 + 3(1.2) = 13.6$ $13.6/2 = 6.8$															

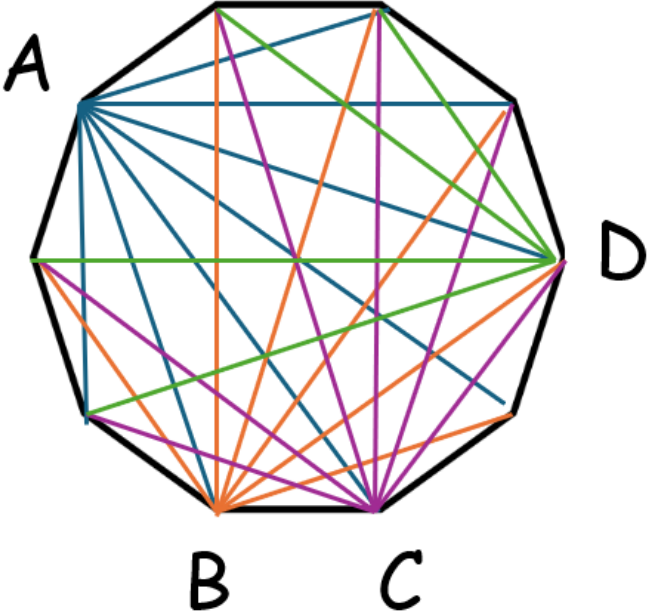
"Math is Cool" Championships -- 2025-26

4th Grade

Team Test Solutions

	Answer	Solution
1	27 [\$]	Mr. Sampson sells 6 watermelons for \$4.50 each. As an integer amount of dollars, how much money did he make? He gained $(\$4.50 \times 6) = \27
2	60 [degrees]	In the regular hexagon shown here, what is the measurement of the angle marked with the question mark, in degrees? A regular hexagon has 120 degrees in each corner. Half of 120 degrees is 60 degrees. 
3	4 [ways]	Selena rolls a fair six-sided die three times. In how many different ways can she roll three numbers in consecutive decreasing order? For example, 12, 11, 10 are in consecutive decreasing order. The decreasing orders are: (6,5,4) , (5,4,3) , (4,3,2) , (3,2,1)
4	360 [pages]	If Kiyon reads 2 pages per minute, how many pages does he read in 3 hours? $2 \text{ pages} \times 3 \text{ hours} \times 60 \text{ minutes per hour} = 360 \text{ pages}$

5	4 [figure no.]	Which of the following figures numbered from 1 to 5 is NOT a part of the larger figure shown here? Your answer should be the figure number.   1  2  3  4  5 Figure 4 is missing the oval from the bottom right area of the original figure. 
6	250 [\$]	There is a stack of eight cards, numbered one through eight, that has been shuffled and placed face down. The game is to turn over cards one at a time until the sum of any two of them is nine, at which point the game is over. If you get \$50 dollars for each card that you turn over, what is the maximum number of dollars that you can win? Your answer should be an integer amount of dollars. The pairs that sum to 9 are 1+8, 2+7, 3+6, 4+5. Therefore, if you turn over 1, 2, 3, and 4 first, then on your 5th turn you will definitely have a sum of 9. So, 5 cards maximum = \$250.

7	75 [%]	Mateo wants to split a pie evenly between himself and his four friends, but two of his friends want pieces that are half the size of the rest. What percent of the pie do Mateo and his friends who have full pieces eat altogether? The pie is split into five parts, with three parts of $x=1$ value and two of $x=\frac{1}{2}$ value. This is the same as 4 slices with one slice split in two. There are 3 out of 4 slices that are not split, or 75%.
8	525 [minutes]	Ada bikes a 42-kilometer race in $3\frac{1}{2}$ hours at a constant pace. Harper rides the first half of the distance at $\frac{1}{2}$ of Ada's pace and the second half of the distance at $\frac{1}{3}$ of Ada's pace. How long, in minutes, does it take Harper to ride the entire race? Ada's pace: $42 \text{ km} / 3.5 \text{ hrs} = 12 \text{ km/hr}$ $21 \text{ km} / 6 \text{ km/hr} = 3.5 \text{ hrs}$ $21 \text{ km} / 4 \text{ km/hr} = 5.25 \text{ hrs}$ $8.75 \text{ hrs} * 60 \text{ mins/hr} = 525 \text{ mins}$
9	23 [diagonals]	A decagon has 4 vertices labeled A, B, C, and D. If a diagonal is drawn from each labeled vertex to all other vertices (labeled and unlabeled), how many total distinct diagonals are drawn?  From A, you can draw 7 diagonals. From B, you can draw 7 minus the diagonal already drawn to A. So 6. From C you can draw 7 minus the diagonal already drawn to A. So 6. From D you can draw 7 minus the diagonals already drawn to A, B, and C. So 4. $7 + 6 + 6 + 4 = 23$ diagonals

10

4 [= 2026th term]

In a particular sequence of numbers, the first term is 48. The rest of the terms are determined by these rules:

1. If a term is even, then divide it by 2 to get the next term.
2. If a term is odd, then multiply it by 3 and add 1 to get the next term.

What is the 2026th term in the sequence?

The sequence will start as follows:

Term #	Term
1	48
2	24
3	12
4	6
5	3
6	10
7	5
8	16
9	8
10	4
11	2
12	1
13	4
14	2
15	1
16	4
17	2
18	1
...	...

Notice that the terms begin to repeat in a pattern of 4, 2, 1, ..., where '1' coincides with multiples of 3. 2026 is not a multiple of 3: $2026/3 = 675r1$, therefore it will be at the beginning of the 4, 2, 1 sequence with a term of 4.

"Math is Cool" Championships -- 2025-26

4th Grade

Linda Moore Triple Jump Solutions

	Answer	Solution
1	9	In the following correct equations, the same number goes in both circles. What number goes in the square? $\square \times 3 = \bigcirc$ $\bigcirc + 12 = 39$ $39 - 12 = 27$ (circle number) $27 \div 3 = 9$
2	300 [cm]	Three meters is equal to how many centimeters? $3 \text{ m} \times 100 \text{ cm/m} = 300$
3	34 [m]	Ivan has a rectangular garden with a length of 8 meters and a width of 5 meters. He wants to put a rectangular fence around the garden that is 1 meter away from the garden on all sides. What is the perimeter of the fence, in meters? The perimeter of the garden is 26. Adding a fence 1 meter away means adding 2 meters of perimeter on each corner. $2 \times 4 = 8$ $26 + 8 = 34$
4	64	Diego is trying to guess what number Leonard is thinking of. Leonard gives him three clues: 1. It is a two-digit whole number. 2. The sum of its digits is 10. 3. It is a square number. What number is Leonard thinking of? Two digit numbers whose digits add up to 10 are: 19, 28, 37, 46, 55, 64, 73, 82, 91. 64 is the only square
5	90	What is the next number in this sequence? 102, 99, 96, 93, ... Pattern is: -3 each time $93 - 3 = 90$

6	30 [marbles]	Gavin has a collection of marbles. He gives 5 marbles to his friend Glen. After recounting his marbles, he notices he still has 4 times as many marbles left as the number he gave away. How many marbles does Gavin need to buy to end up with twice his original amount? Gave away 5. Still has $4 \times 5 = 20$, so he originally had 25. Twice the original amount = $25 \times 2 = 50$. Needs to buy: $50 - 20 = 30$
7	36 [cookies]	Tyson baked a big batch of cookies. He gave $\frac{1}{3}$ of the cookies to his neighbors. Then, he ate $\frac{1}{4}$ of the remaining cookies himself. After that, Tyson gave 6 cookies to his best friend. Finally, Tyson decided to save half of the cookies he had left for later. He now has exactly 6 cookies left. How many cookies did Tyson bake at the very start? $6 \times 2 = 12$ $12 + 6 = 18$ $18 + \frac{1}{3}(18) = 24$ $24 + \frac{1}{2}(24) = 36$
8	7 [different scores]	In the game Flip Flap Floop, a player can earn 3, 7 or 11 points in a single turn. How many different total scores are possible for a single player after three turns? There are a total of $3 \times 3 \times 3 = 27$ different ways to get scores after 3 turns. However, a lot of them are repeats. For example, 3 3 7, 3 7 3 and 7 3 3 will all give a total of 13 points for 3 turns. The unique totals are: $3 + 3 + 3 = 9$ $3 + 3 + 7 = 13$ $3 + 7 + 7$ OR $3 + 3 + 11 = 17$ $7 + 7 + 7$ OR $3 + 7 + 11 = 21$ $3 + 11 + 11 = 25$ $7 + 11 + 11 = 29$ $11 + 11 + 11 = 33$

9	45 [minutes]	Ishanna is watering her garden using a 9-liter watering can. Ishanna fills her watering can at a rate of 3 liters every 150 seconds. She waters her plants using the watering can, pouring 2 liters every 5 minutes. If Ishanna starts with an empty can and needs 13.5 liters of water to water her garden, what is the shortest amount of time, in minutes, she needs to water her garden? She does not have to fill the can completely if it is not necessary. Ishanna needs to fill a full can, empty a full can, then fill half a can, and empty half a can. The total time will be 1.5x the time it takes to fill one can plus 1.5x the time it takes to empty one can, or 1.5x the time it takes to fill and empty one can. Fill: $9 / 3 = 3$ $3 * 150 = 450$ $450 / 60 = 7.5$ minutes Empty: $9 / 2 = 4.5$ $4.5 * 5 = 22.5$ minutes $22.5 + 7.5 = 30$ $1.5 * 30 = 45$
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10 **15 [= sum]**

In the following crossword puzzle, every answer is a whole number, and exactly one digit goes in each box. Only the digits 1 through 9 are used, and they can be used more than once.

Once you have solved the puzzle, find the sum of the digits that are in the boxes numbered

3, 4, 8 and 9. What is the sum?

Across clues:

- 1 Across: A multiple of 7
- 3 Across: A prime number
- 5 Across: Consecutive ascending (getting bigger) digits
- 7 Across: Equals 10 Across plus 700
- 8 Across: Five different odd numbers
- 10 Across: Equals 11 Across plus 10
- 11 Across: A multiple of 5

Down clues:

- 1 Down: A multiple of 11
- 2 Down: Five different odd numbers
- 3 Down: A palindrome equal to 8 Across minus 2 Down
- 4 Down: The square of an even square
- 6 Down: The first and last digits add up to the middle digit
- 8 Down: A multiple of 11
- 9 Down: A multiple of 5

1 2	2 1		3 6	4 1
5 2	3 3	6 4	5 5	6 6
	7 7	7 7	5 5	
8 7	9 9	3 3	5 5	9 1
10 7	5 5		11 6	5 5

$$6 + 1 + 7 + 1 = 15$$

“Math is Cool” Championships -- 2025-26

4th Grade

College Bowl Round #1 Solutions

#	Problem	Answer
1	What number comes next in the pattern that starts as follows: two, five, eight, eleven, and so on? +3 every time	14
2	A set of numbers is as follows: five, four, two, six, and four. What is the mode of this set? 4 is repeated the most	4 [= mode]
3	What is the greatest common factor of twelve and eighteen? 12: 1, 2, 3, 4, 6, 12 18: 1, 2, 3, 6, 9, 18	6 [= gcf]
4	Rosie has a bag with six red marbles, six green marbles, and eight blue marbles. If she draws one marble from the bag, what is the probability in percent that the marble is red or green? $6 + 6 = 12$ $12 / 20 = 60\%$	60 [%]
5	A water tank can hold two-hundred forty gallons of water. If one- hundred thirty-five gallons of water are already in the tank, and forty-five gallons are then removed, how many gallons should be added to the tank to make it full? $135 - 45 = 90$ $240 - 90 = 150$	150 [gallons]
6	Brecken has one quarter, three dimes, and one nickel. How many more nickels does Brecken need to make one dollar? $25 + 30 + 5 = 60$ $100 - 60 = 40$ $40 / 5 = 8$ nickels	8 [nickels]
7	A rectangle has a length of twelve centimeters and a width that is one-third the length. What is its perimeter in centimeters? Width = $1/3(12) = 4$ $12 + 4 + 12 + 4 = 32$.	32 [cm]
8	How many prime numbers are less than thirteen? 2, 3, 5, 7, 11	5 [prime nos.]

9	If a clock shows the time seven-oh-seven, what number on the clock face will the minute hand land on in sixty-three minutes? 63 minutes after 7:07 is 8:10. Ten minutes is shown by the number 2.	2
10	You can read two pages every three minutes. How many pages can you read in one hour? $60 / 3 = 20$ $20 * 2 = 40$	40 [pages]

“Math is Cool” Championships -- 2025-26

4th Grade

College Bowl Round #2 Solutions

#	Problem	Answer
1	What is the median of the following four numbers: five, seven, seventeen, four? In order are 4, 5, 7, 17. 6 is between 5 and 7	6 [= median]
2	What is the smallest prime number greater than ten? 11 is the smallest prime number greater than 10	11
3	If yesterday was a Tuesday and it was the eighth day of the month, how many days will it be until the fourth Tuesday of the month? Tuesday the 8th is the second Tuesday. If the second Tuesday was yesterday, the fourth Tuesday will be in 13 days.	13 [days]
4	What is one-fifth of one-half of seventy? $\frac{1}{2} * \frac{1}{5} * 70 = 7$	7
5	A right triangle has two legs of lengths three inches and four inches. What is the area of the triangle in square inches? $3 * 4 = 12$ $12 / 2 = 6$	6 [sq. inches]
6	A spinner has four equal sections with colors green, red, blue, and yellow. If you spin it sixteen times, how many total times do you expect it to land on either red, blue, or yellow? Red, blue, and yellow make up 3 of the 4 sections. $\frac{3}{4}$ of 16 is 12.	12 [times]
7	Fourteen cats are in a room. Two cats enter and one cat leaves the room, then three cats enter and two cats leave the room. How many cats are now in the room? $14 + 2 - 1 + 3 - 2 = 16$	16 [cats]
8	Mishka drinks three cups of water every hour. How many hours does it take Mishka to drink three quarts of water? 3 quarts is 12 cups. $12 \text{ cups} / 3 \text{ cups / hour} = 4 \text{ hours}$	4 [hours]
9	A sequence of numbers starts with three and doubles each time. What is the fourth number in the sequence? 3, 6, 12, 24	24 [= 4 th no.]

10	X plus two equals seven. What is X minus four? $X + 2 = 7$ $X = 5$ $5 - 4 = 1$	1 [= x - 4]
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“Math is Cool” Championships -- 2025-26

4th Grade

College Bowl Round #3 Solutions

#	Problem	Answer
1	Sophie starts counting down by fives, starting at fifty-two. What is the fifth number she names? 52, 47, 42, 37, 32	32 [= 5 th no.]
2	For every five stuffed animals a company makes and sells, they profit twenty dollars. How many dollars do two stuffed animals profit? $20 \div 5 = 4$ $4 * 2 = 8$	8 [\$]
3	Stephanie is thinking of a number. It is greater than fifteen, less than thirty, and a multiple of eleven. What is the number? It is the only number between 15 and 30 that is a multiple of 11.	22
4	What is the sum of all the odd whole numbers less than ten? $1+3+5+7+9=25$	25 [= sum]
5	A group of eight friends wants to share some candy equally between them. If they have seventy-seven pieces of candy, and each friend gets an equal number, how many pieces are left over? $77 \div 8 = 9 \text{ r } 5$	5 [pieces]
6	If the average of three numbers is twenty, and two of the numbers are twenty-six and fifteen, what is the third number? All three numbers must add up to $20*3$, or 60. $60 - 26 - 15 = 19$	19 [= 3 rd no.]
7	A rectangle has a perimeter of thirty feet and a width of five feet. What is its area in square feet? $30 = 5 + 5 + 2x$ $2x = 20$ $x = 10$ $10 * 5 = 50$	50 [square feet]
8	A ten-sided die is numbered zero through nine. On a single roll, what is the probability in percent of rolling a number less than four? Outcomes: 0, 1, 2, 3 are less than 4. $4/10 = 40\%$	40 [%]

9	A car travels five miles in ten minutes. How many miles will it travel in one hour? $60 / 10 = 6$ $5 * 6 = 30$	30 [miles]
10	How many even whole numbers are there between twenty-one and thirty-five? 22, 24, 26, 28, 30, 32, 34	7 [even whole nos.]

“Math is Cool” Championships -- 2025-26

4th Grade

College Bowl Round #4 Solutions

#	Problem	Answer
1	What is the next number in the sequence that begins as follows: three, four, six, nine, and so on? Adding 1 more each time $9+4=13$	13
2	How many total factors does the number twenty-four have? 1, 2, 3, 4, 6, 8, 12, 24	8 [total factors]
3	In a group of ladybugs, each ladybug has ten spots. What is the fewest possible number of ladybugs needed to have a total of at least one-hundred twenty-one spots? The smallest number that is larger than 121 and divisible by 10 is 130. 13 ladybugs are needed to get 130 spots.	13 [ladybugs]
4	A movie theater has twelve rows of seats with eighteen seats in each row. One show sold one hundred sixty-eight tickets. How many empty seats were there? $12 \times 18 = 216$ $216 - 168 = 48$ seats	48 [seats]
5	Quentin is going to fill a box with measurements forty by forty by sixty units, using identical cubes. What is the minimum number of cubes needed? If the cubes are 20x20 units, a 2x3 array will fill the box.	12 [cubes]
6	Felix has a bookshelf with forty-five books of four types: Nonfiction, fantasy, biography, and science fiction. If there are fourteen nonfiction books, sixteen fantasy books, and three biography books, how many science fiction books are on the bookshelf? $45 - 14 - 16 - 3 = 12$	12 [books]
7	X minus three equals ten minus two. What is X? $x - 3 = 8$ $x = 11$	11 [= x]
8	What is nineteen times four? $19 \times 4 = 76$	76

9	Landon collected three apples on Monday, four apples on Tuesday, and eight apples on Wednesday. What was the average number of apples Landon collected per day? $3+4+8=15$ $15 / 3 = 5$	5 [apples]
10	Ira flips a coin nine times and gets nine heads in a row. If he flips it a tenth time, what is the probability in percent that the coin will land heads? The individual outcome is unaffected by previous outcomes, so it will still be 50%.	50 [%]

“Math is Cool” Championships -- 2025-26

4th Grade

College Bowl Round #5 Solutions

#	Problem	Answer
1	On her first four English tests, Addy scored ninety-one, seventy-seven, ninety-four, and eighty-eight. What is the range of her scores on these tests? $94 - 77 = 17$	17 [= range]
2	A number is greater than ten and less than twenty. It is a multiple of four, and is one greater than a prime. What number is it? 12 and 16 are the only multiples of 4 between 10 and 20. 15 is not prime but 11 is.	12
3	A square garden has an area of one-hundred twenty-one square meters. What is the length of one side in meters? The root of 121 is 11.	11 [m]
4	May is eight years old, Leah is half her age. How many years old will Leah be when May is ten years old? May is 8 and Leah is 4. In 2 years May is 10 and Leah is 6.	6 [years]
5	A train leaves at nine forty-five AM and arrives at its destination at one-twenty PM on the same day. How many minutes did the trip take? $9:45$ to $12:45 = 3\text{hrs} = 180$ minutes From $12:45$ to $1:20$ there are 35 minutes $180 + 35 = 215$ minutes total	215 [minutes]
6	A group of three friends wants to plant thirty-six flowers every hour. At a constant rate, how many minutes should it take one friend to plant one flower? $36 / 3 = 12$ $60 / 12 = 5$	5 [minutes]
7	What is the greatest common multiple of four and six that is less than sixty? 48 is the greatest multiple of both 4 and 6 less than 60.	48 [= gcm]
8	A pattern of numbers is formed by counting up by fives, starting with zero as the first term. What is the eighth number in the pattern? 0, 5, 10, 15, 20, 25, 30, 35	35 [= 8 th no.]

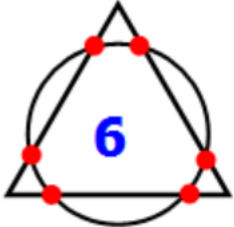
9	How many total sides do three octagons, two hexagons and one trapezoid have? $3 \times 8 + 2 \times 6 + 4 = 24 + 12 + 4 = 40$	40 [sides]
10	What is two-thirds of two hundred seventy? $\frac{2}{3} * 270 = 180$	180

“Math is Cool” Championships -- 2025-26

4th Grade

College Bowl Round #6 Solutions

#	Problem	Answer
1	What is one-thousand five-hundred sixty-five point three five rounded to the nearest ten? 1,565.35 rounds up to 1,570	1,570
2	Javier can eat hot dogs at a rate of four every ten minutes. How many can he eat in twenty-five minutes? $4 * 2.5 = 10$	10 [hot dogs]
3	A rectangular piece of paper has a length of seven inches and an area of forty-two square inches. What is its width in inches? $7 * x = 42$ $42 / 7 = 6$	6 [inches]
4	A box has twenty pencils, five of which are red. If Tanushi takes one pencil without looking, what is the probability in percent that there are fifteen non-red pencils left in the box? There are a total of 15 non-red pencils. For all to remain, a red one must be selected. $5 / 20 = 25\%$	25 [%]
5	Jasmine has forty-one cents in coins, where each coin has a different value. How many coins does she have? $Q + D + N + P = 41$	4 [coins]
6	A farmer has two cows and some chickens. The animals have a total of eighteen legs. How many chickens does the farmer have? $18 - 8 = 10$ $10 / 2 = 5$	5 [chickens]
7	How many times does the digit zero appear in the number one-hundred-one thousand twenty-six? 101,026 2 zeros	2

8	Christine has ten cookies. Her friend gives her some more, doubling the amount Christine has. Christine then gives away eighty percent of her cookies. How many cookies does she have left? $10 * 2 = 20$ $20 - (.8 * 20) = 4$	4 [cookies]
9	What is the units digit when the first five odd whole numbers are multiplied together? $1 \times 3 \times 5 \times 7 \times 9$ Because it includes a 5, but not a 2, it's going to end in 5.	5 [= units digit]
10	What is the maximum number of points of intersection between a triangle and a circle? 	6 [points]